جامعة حفر الباطن
University of Hafr Al Batin

Lecture 15 - Remote Login Protocols (TELNET, SSH)

Objectives

- TELNET Overview.
- TELNET Time-Sharing Environment.
- Network Virtual Terminal (NVT).
- TELNET Embedding.
- TELNET Options.
- SSH Overview.
- SSH Components.
- SSH Port Forwarding.
- SSH Packets

Remote Login Protocols Overview

- The main task of the Internet and its TCP/IP protocol suite is to provide services for users.
- Although there are some specific client/server programs that we discuss before, it would be impossible to write a specific client-server program for each demand.
- The better solution is a general-purpose client-server program that lets a user access any application program on a remote computer.

Remote Login Protocols Overview

- in other words, allow the user to log on to a remote computer.
- After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.
- In this chapter, we discuss two of these application programs:
  - TELNET.
  - SSH.

TELNET Protocol

TELNET

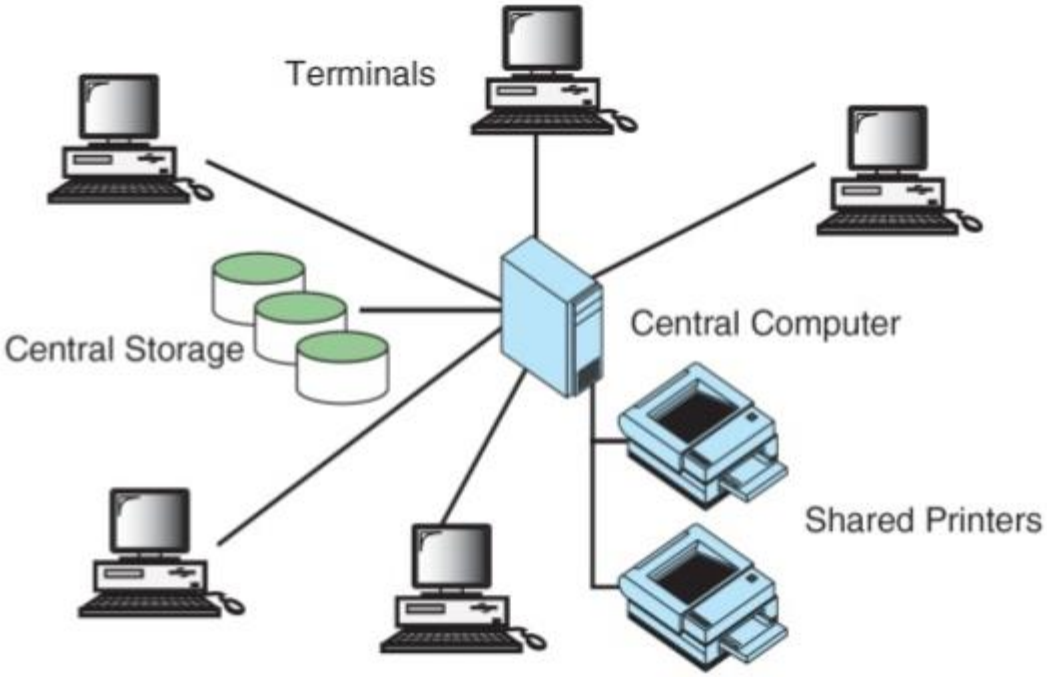
- **TELNET:** is the standard TCP/IP protocol for virtual terminal service as proposed by ISO.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.

- TELNET is a general-purpose client-server application program.
- TELNET is an abbreviation for TErminaL NETwork.

TELNET & Time-Sharing Environment

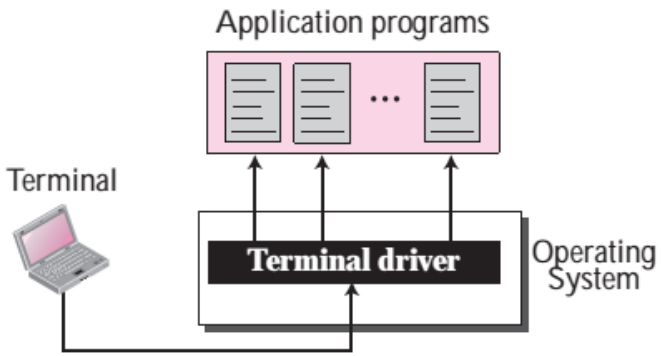
- TELNET was designed at a time when most operating systems, such as UNIX, were operating in a time-sharing environment.
- **Time-Sharing Environment means that a large computer supports multiple users.**
  - The interaction between a user and the computer occurs through a terminal, which is usually a combination of keyboard, monitor, and mouse.
  - All of the processing must be done by the central computer.
  - When a user types a character on the keyboard, the character is usually sent to the computer and echoed to the monitor.
- Time-sharing creates an environment in which each user has the illusion of a dedicated computer.

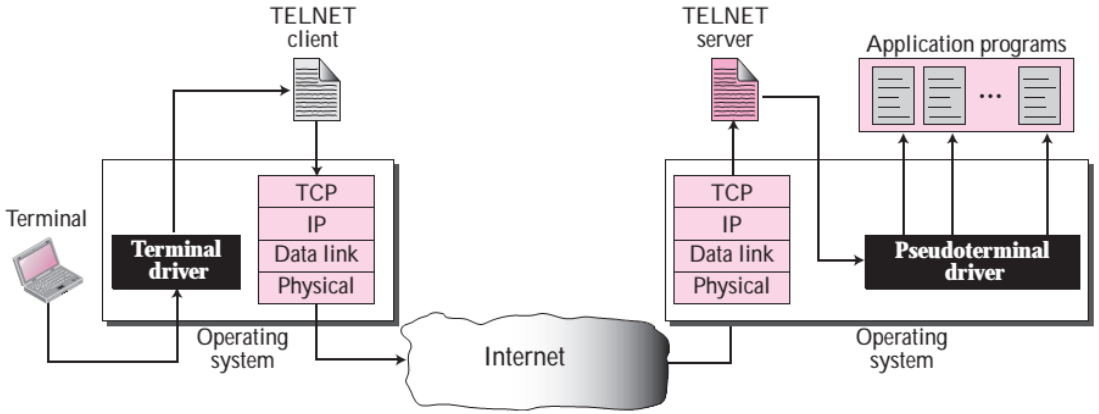
TELNET & Time-Sharing Environment



TELNET & Time-Sharing Environment

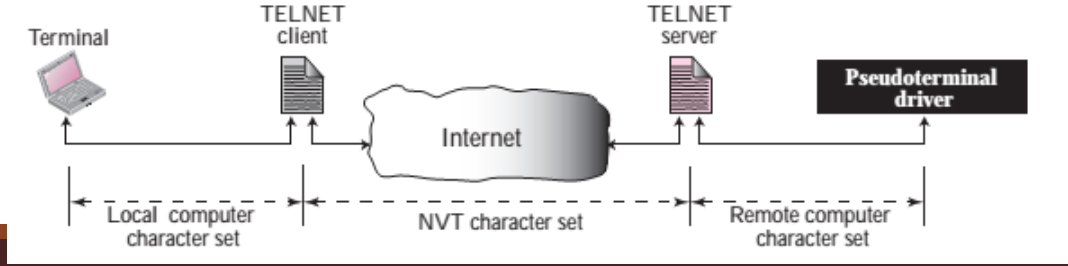
- In a time-sharing environment, users are part of the system with some right to access resources.
  - Each authorized user has an identification and probably a password.
- There are *two types of login in time-sharing systems*:
  - **Local Login:** when a user logs into a local time-sharing system.
  - **Remote Login:** when a user wants to access an application program or utility located on a remote machine.





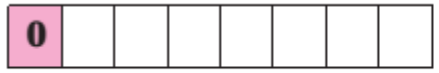
Network Virtual Terminal (NVT)

- The mechanism to access a remote computer is complex.
  - This is *because* every computer and its operating system accepts a special combination of characters as tokens.
  - For example, the end-of-file token (Ctrl+z in DOS OS, Ctrl+D in UNIX OS).
- TELNET solves this problem by defining a universal interface called the Network Virtual Terminal (NVT) character set.
  - the client TELNET translates characters (data or commands) coming from the local terminal into NVT form and delivers them to the network.
  - The server TELNET translates data and commands from NVT form into the form acceptable by the remote computer.

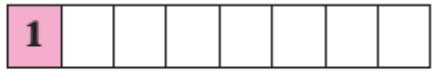


Network Virtual Terminal (NVT)

- NVT uses two sets of characters, one for data and one for control.
  - Both are 8-bit bytes.
  - Data Characters: the highest order bit is set to 0.
  - Control Characters: the highest order bit is set to 1.



a. Data Character



b. Control Character

Network Virtual Terminal (NVT)

- Some NVT control characters.

Character	*Decimal*	*Binary*	*Meaning*
EOF	236	11101100	End of file
EOR	239	11101111	End of record
SE	240	11110000	Suboption end
NOP	241	11110001	No operation
DM	242	11110010	Data mark
BRK	243	11110011	Break
IP	244	11110100	Interrupt process
AO	245	11110101	Abort output
AYT	246	11110110	Are you there?
EC	247	11110111	Erase character
EL	248	11111000	Erase line
GA	249	11111001	Go ahead
SB	250	11111010	Suboption begin
WILL	251	11111011	Agreement to enable option
WONT	252	11111100	Refusal to enable option
DO	253	11111101	Approval to option request
DONT	254	11111110	Denial of option request
IAC	255	11111111	Interpret (the next character) as control

TELNET Embedding

- TELNET uses only one TCP connection.
  - The server uses the well-known port 23 and the client uses an ephemeral port.
  - The same connection is used for sending both data and control characters.
- TELNET accomplishes this by embedding the control characters in the data stream.
  - To distinguish data from control characters, each sequence of control characters is preceded by a special control character called interpret as control (IAC).

TELNET Embedding

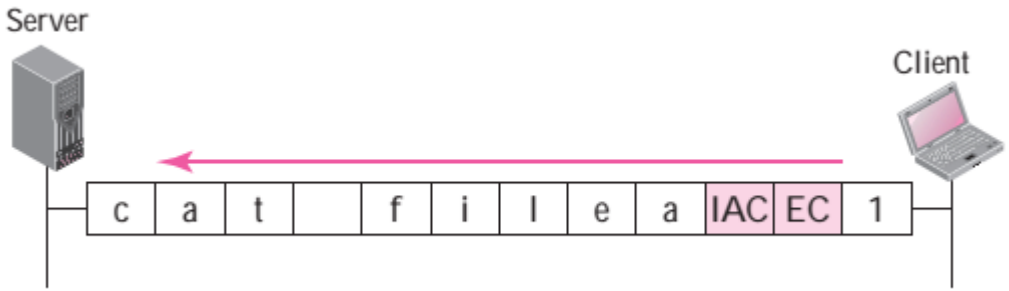
- Example:
  - Imagine a user wants a server to display a file (file1)

cat file1

  - However, the name of the file has been mistyped (filea instead of file1).

cat filea<backspace>1

  - However, in the default implementation of TELNET, the user cannot edit locally.
    - The editing is done at the remote server.
  - The backspace character is translated into two remote characters (IAC EC).



TELNET Options

- TELNET lets the client and server negotiate options before or during the use of the service.
- Options are extra features available to a user with a more sophisticated terminal.
  - Users with simpler terminals can use default features.

Code	*Option*	*Meaning*
0	Binary	Interpret as 8-bit binary transmission
1	Echo	Echo the data received on one side to the other
3	Suppress go-ahead	Suppress go-ahead signals after data
5	Status	Request the status of TELNET
6	Timing mark	Define the timing marks
24	Terminal type	Set the terminal type
32	Terminal speed	Set the terminal speed
34	Line mode	Change to line mode

TELNET Options

- **Binary:** allows the receiver to interpret every 8-bit character received, except IAC, as binary data.
- **Echo:** allows the server to echo data received from the client.
  - In this case, the user terminal usually does not echo characters when they are typed but waits until it receives them from the server
- **Suppress go-ahead:** suppresses the go-ahead (GA) character.
- **Status:** allows the user or the process running on the client machine to get the status of the options being enabled at the server site.
- **Timing mark:** allows one party to issue a timing mark that indicates all previously received data has been processed.

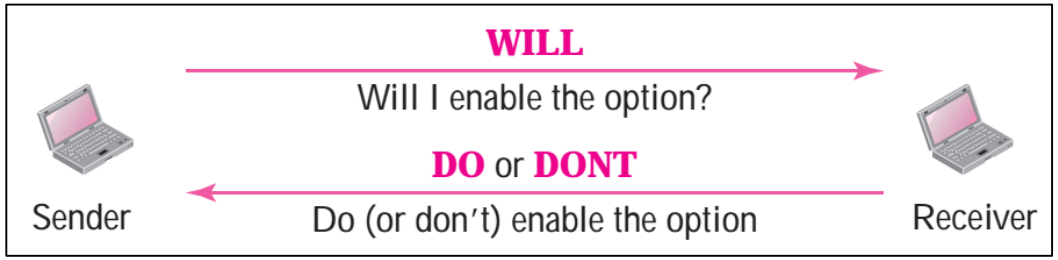
TELNET Options

- **Terminal type:** allows the client to send its terminal type.
- **Terminal speed:** allows the client to send its terminal speed.
- **Line mode:** allows the client to switch to the line mode.
- To use any of the options, first requires ***option negotiation*** between the client and the server. Four control characters are used for this purpose:

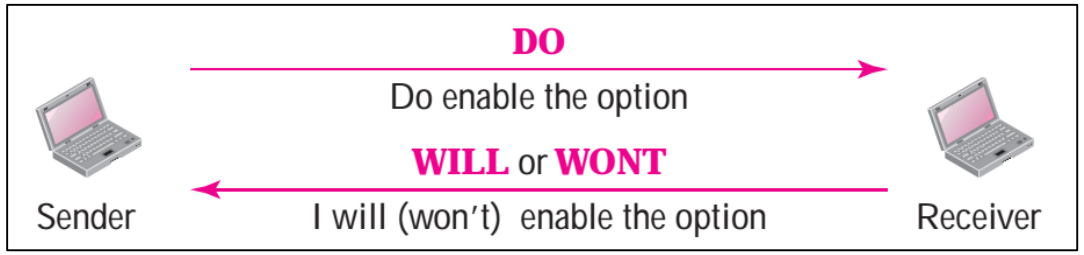
Character	*Code*	*Meaning 1*	*Meaning 2*	*Meaning 3*
WILL	251	Offering to enable	Accepting to enable	
WONT	252	Rejecting to enable	Offering to disable	Accepting to disable
DO	253	Approving to enable	Requesting to enable	
DONT	254	Disapproving to enable	Approving to disable	Requesting to disable

TELNET Options

- Option negotiation: Offer to enable an option

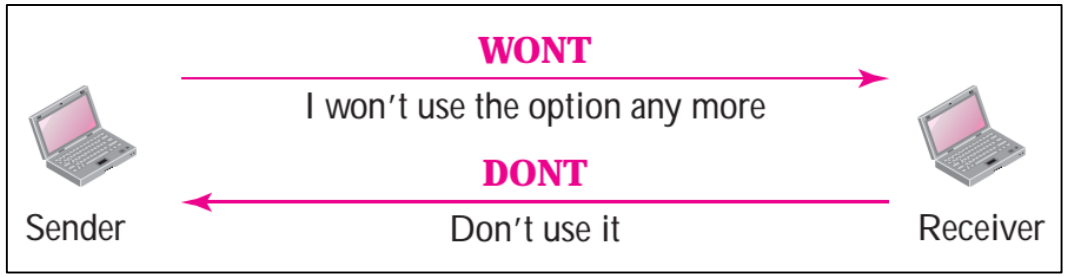


- Option negotiation: Request to enable an option

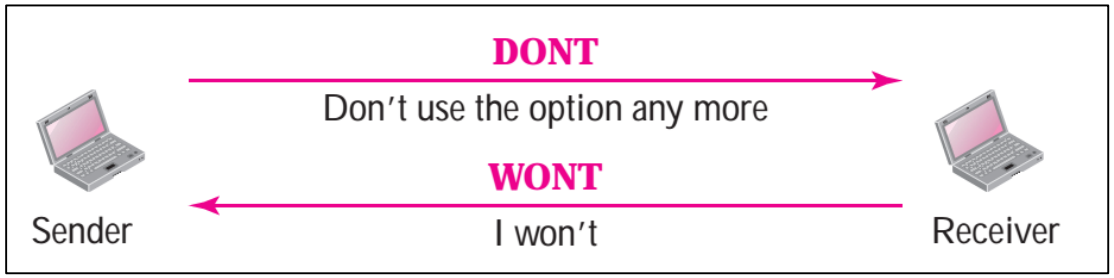


TELNET Options

- Option negotiation: Offer to disable an option

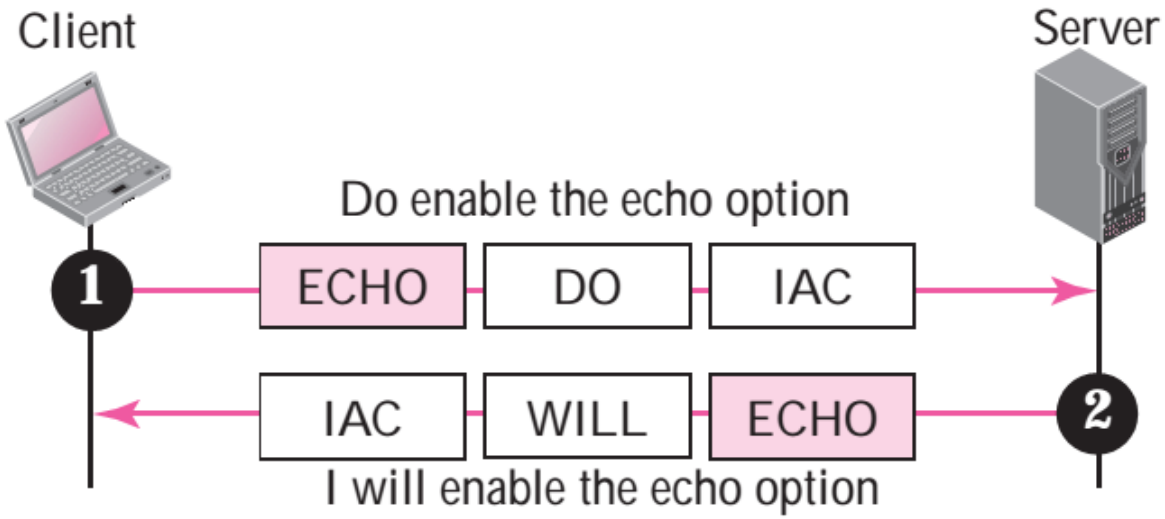


- Option negotiation: Request to disable an option



TELNET Options

- Option negotiation: Example:
  - The client wants the server to echo each character sent to the server.

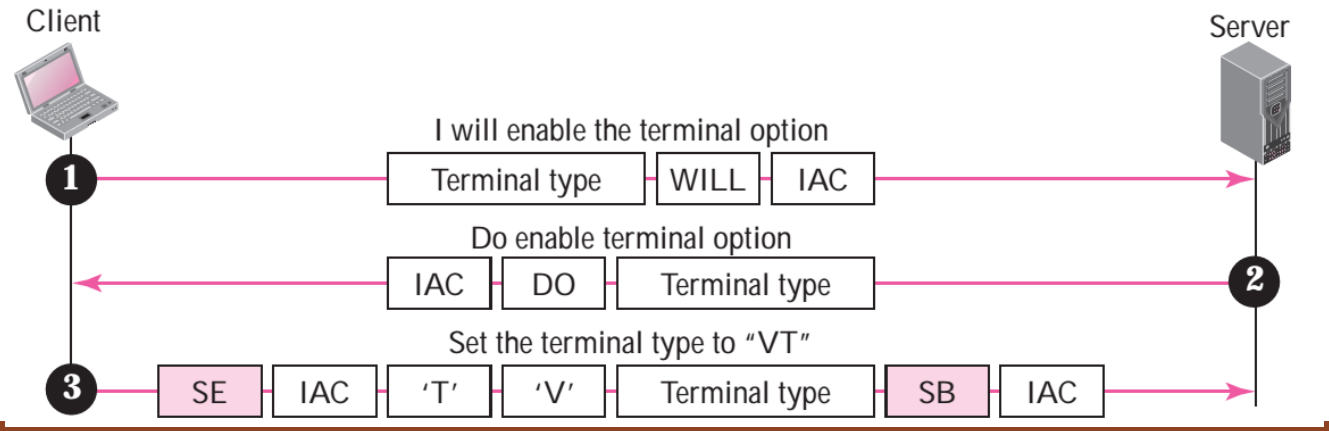


TELNET Options

Suboption Negotiation:

- Some options require additional information.
  - For example, to define the type or speed of a terminal, the negotiation includes a string or a number to define the type or speed.

Character	*Decimal*	*Binary*	*Meaning*
SE	240	11110000	Suboption end
SB	250	11111010	Suboption begin



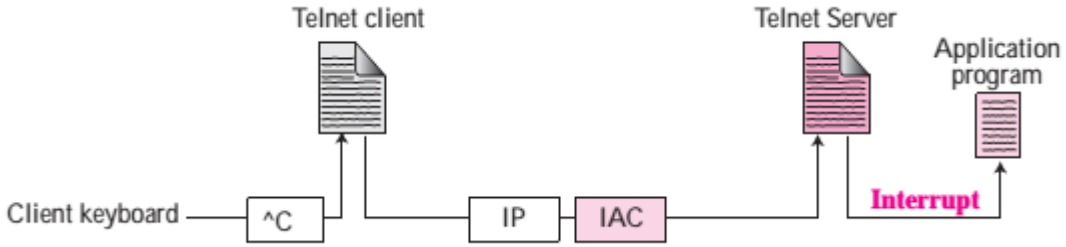
Controlling the Server

- Some control characters can be used to control the remote server.
  - When an application program is running on the local computer, special characters are used to interrupt (abort) the program (for example, Ctrl+c), or erase the last character typed (for example, delete key or backspace key), and so on.
  - However, when a program is running on a remote computer, these control characters are sent to the remote machine.
- The user still types the same sequences, but they are changed to special characters and sent to the server.

Character	*Decimal*	*Binary*	*Meaning*
IP	244	11110100	Interrupt process
AO	245	11110101	Abort output
AYT	246	11110110	Are you there?
EC	247	11110111	Erase the last character
EL	248	11111000	Erase line

Controlling the Server

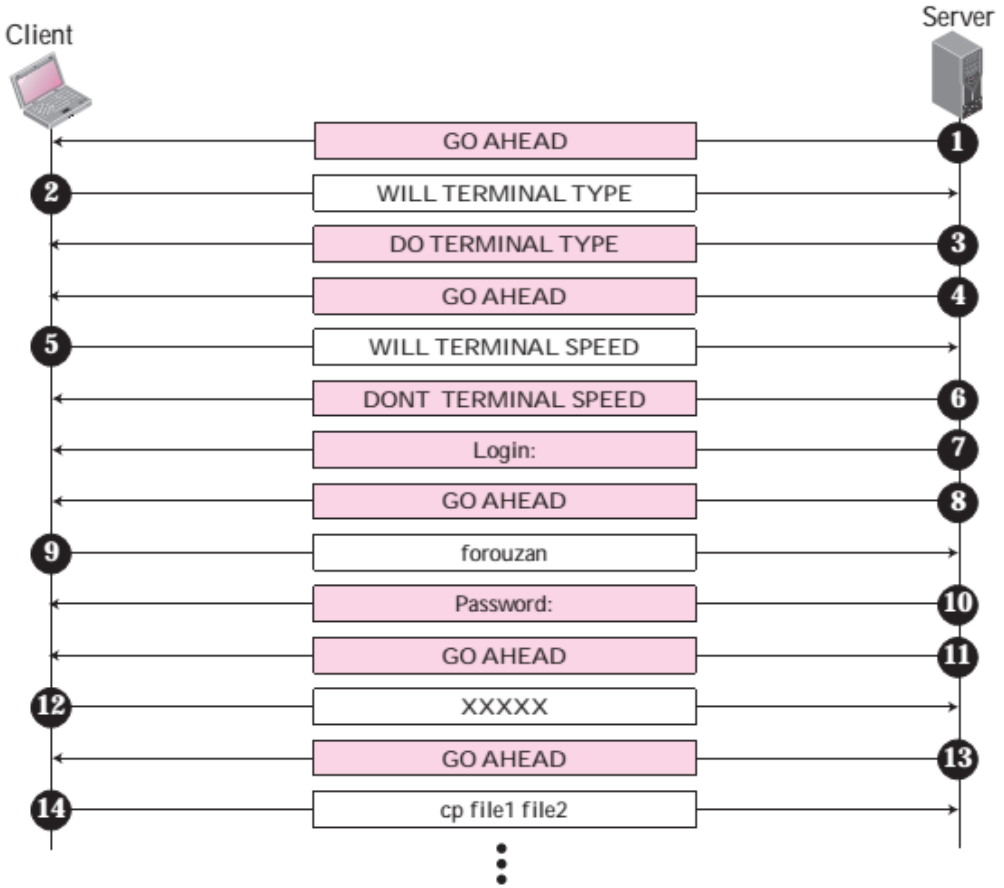
- Some of the characters that can be sent to the server to control the application program that is running there.
  1. **IP (interrupt process):** to interrupt (abort) the program.
  2. **AO (abort output):** to allows the process to continue without creating output.
  3. **AYT (are you there?):** to determine if the remote machine is still up and running.
  4. **EC (erase character):** to erase the last character typed.
  5. **EL (erase line):** erase the current line in the remote host.



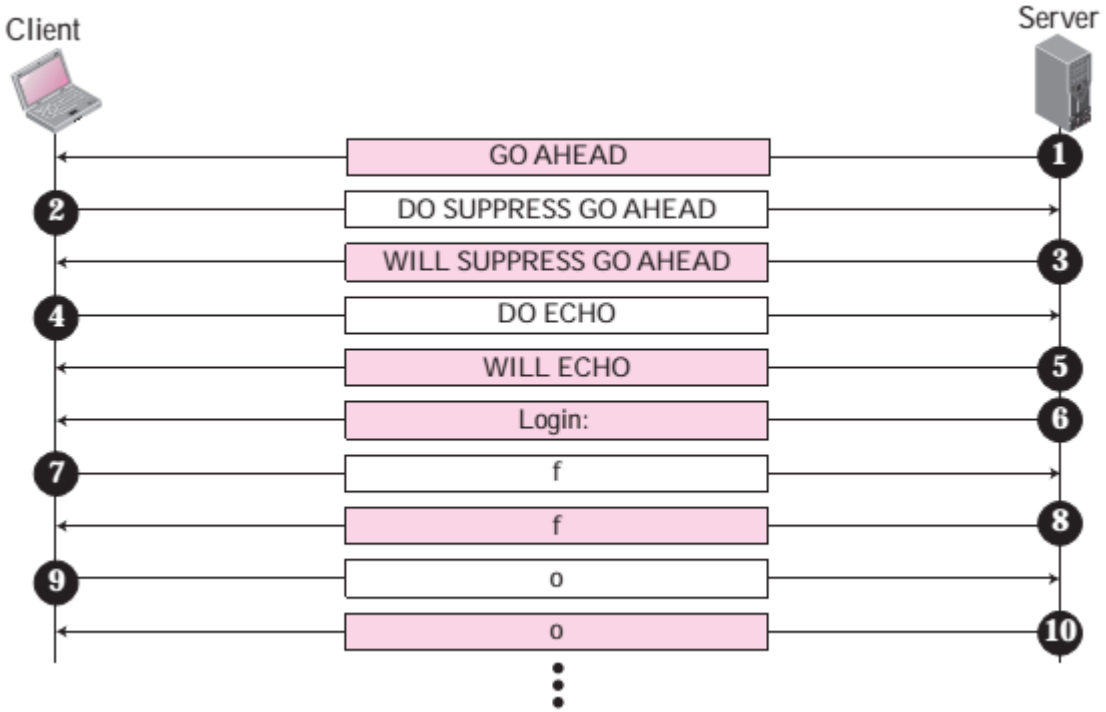
TELNET Modes

- Most TELNET implementations operate in one of THREE modes:
  1. **Default mode.**
     - Echoing to the screen is done by the client.
     - Client does not send characters until a whole line is completed.
     - Client waits for the GA (go ahead) command from the server before accepting a new line from the user.
     - Con: The operation is half-duplex (not efficient).
  2. **Character mode.**
     - Each character typed is sent by the client to the server.
     - Server normally echoes the character back to be displayed on the client screen.
     - Con: Echoing of the character can be delayed if the transmission time is long.
     - Con: It creates overhead (traffic) for the network

TELNET Modes



Default Mode



Character Mode

TELNET Modes

3. **Line mode.**
   - Has been proposed to compensate for the deficiencies of the previous modes.
   - Line editing is done by the client.
     - echoing, character erasing, line erasing, and so on…
   - Client sends the whole line to the server.
   - It is full-duplex
     - The client sending one line after another, without the need for an intervening GA (go ahead) character from the server.

TELNET User Interface

- The normal user does not use TELNET commands as defined above.
- Usually, the operating system (UNIX, for example) defines an interface with user-friendly commands.

Command	*Meaning*	*Command*	*Meaning*
open	Connect to a remote computer	set	Set the operating parameters
close	Close the connection	status	Display the status information
display	Show the operating parameters	send	Send special characters
mode	Change to line or character mode	quit	Exit TELNET

TELNET Security

- TELNET suffers from security problems.
- Although TELNET requires a login name and password (when exchanging text), often this is not enough.
- A microcomputer connected to a broadcast LAN can easily eavesdrop using snooper software and capture a login name and password (even if it is encrypted).

SSH Protocol

SSH

- Another popular remote login application program is **Secure Shell (SSH).**
- SSH, like TELNET, uses TCP as the underlying transport protocol, but:
  - SSH is more secure.
  - SSH provides more services than TELNET.

SSH Versions

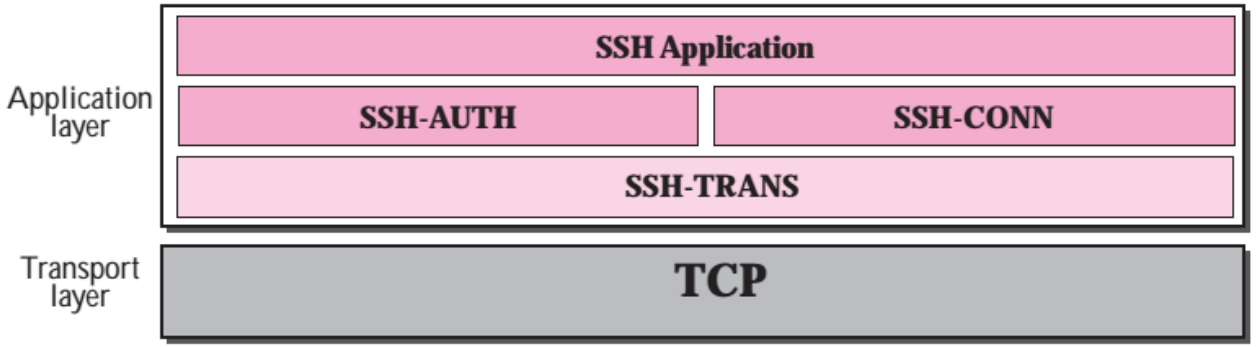
- There are two versions of SSH: SSH-1 and SSH-2.
  - They are incompatible.
  - The first version, SSH-1 is now deprecated because of security flaws in it.
- In this lecture, we discuss only SSH-2.

SSH Components

SSH has four components:

1. SSH Transport-Layer Protocol (SSH-TRANS):

- TCP is not a secured transport layer protocol.
- SSH first uses this independent protocol to creates a secured channel on top of TCP.
- It does the following:
  1. Establish a secure connection between the client and the server
  2. Authenticate the server.

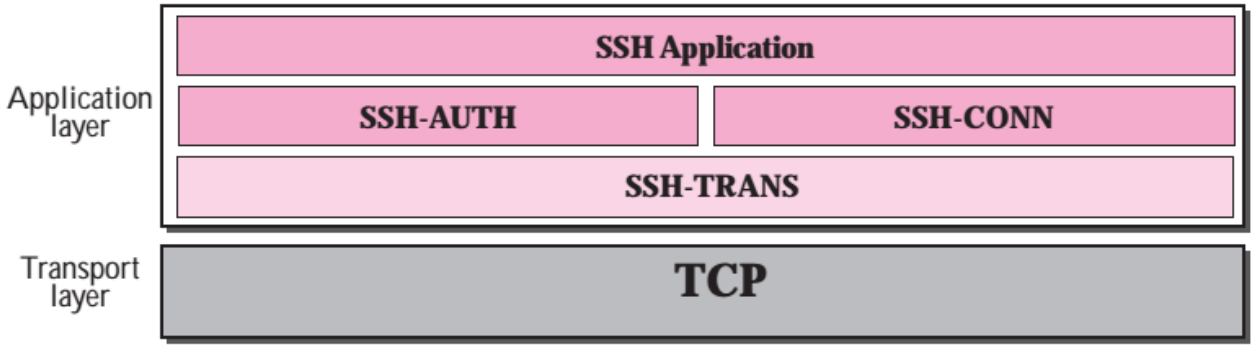


SSH Components

SSH has four components:

2. SSH Authentication Protocol (SSH-AUTH):

- After a secure channel is established between the client and the server and the server is authenticated for the client.
- SSH can call another software that can authenticate the client for the server

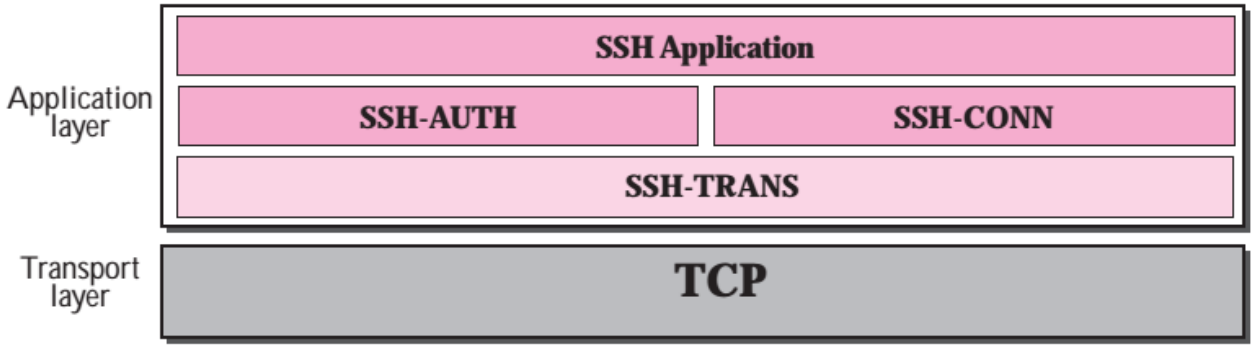


SSH Components

SSH has four components:

3. SSH Connection Protocol (SSH-CONN):

- After the secured channel is established and both server and client are authenticated for each other.
- SHH can call this protocol.
- One of the services provided by the SSH-CONN protocol is to do multiplexing.
- ISSH-CONN lets the client create multiple logical channels over it.

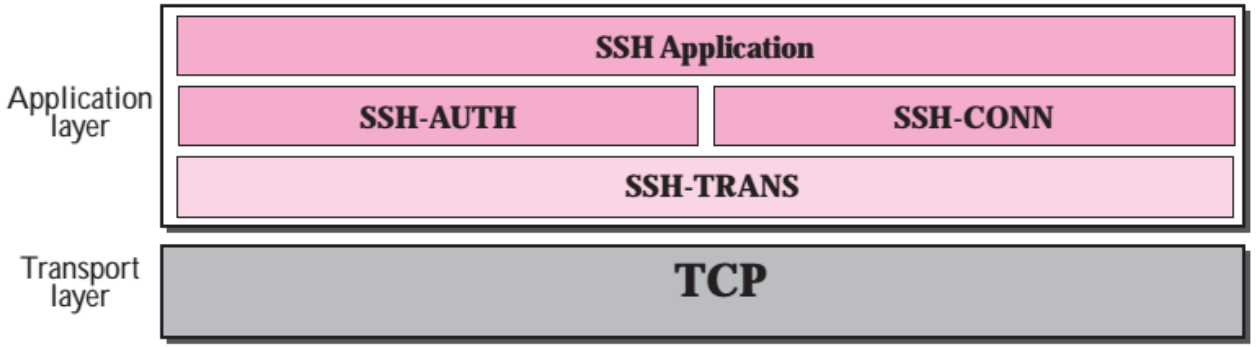


SSH Components

SSH has four components:

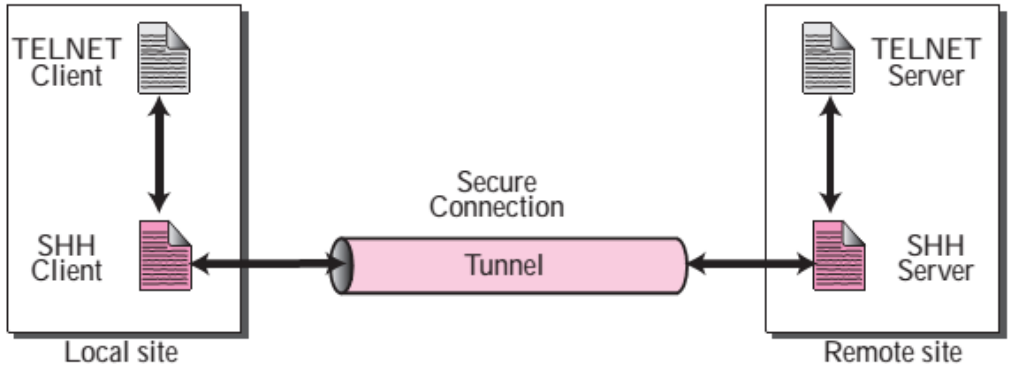
4. SSH Applications:

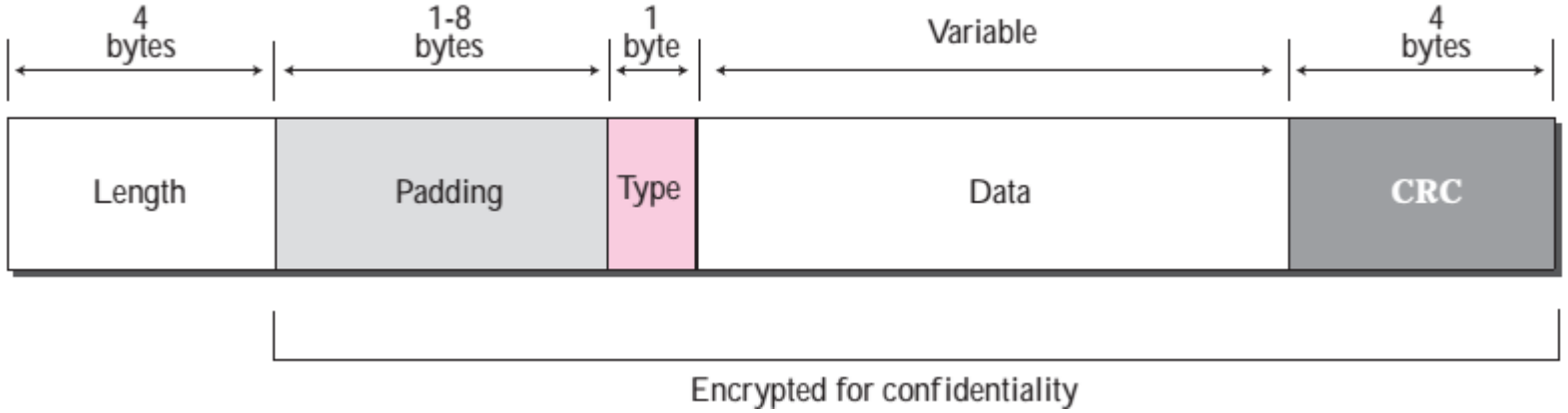
- After the connection phase is completed, SSH allows several application programs to
- use the connection.
- Each application can create a logical channel as described above and then benefit from the secured connection.
  - Remote login applications.
  - File transfer applications.



SSH Port Forwarding

- One of the services provided by the SSH is to provide port forwarding.
- We can use the secured channels available in SSH to access an application program that does not provide security services.
  - Application such as TELNET and SMTP.
- SSH port forwarding creates a tunnel through which the messages belonging to other protocol can travel.
  - sometimes referred to as **SSH tunneling**.



SSH Packets

- SSH Packet Fields:
  1. **Length:** the length of the packet including the type, the data, and the CRC field, but not the padding and the length field.
  2. **Padding:** added to packets to make attacks on the security provision more difficult.
  3. **Type:** the type of the packet used by SSH protocols.
  4. **Data:** variable length (can be found be deducting 5 from Length value).
  5. **CRC:** (cyclic redundancy check) used for error detection.